

Mateusz Kapusta

EDUCATION

*Master of Science, **2023-2026** (expected)*

Master of Science in Astrophysics

University of Warsaw, Faculty of Physics, Poland

*Bachelor of Science, **2020-2023***

Inter-faculty Individual Studies in Mathematics and Natural Sciences, University of Warsaw, Poland

Grade: 4.96/5 (2-5 scale, 5 is highest), graduated with distinctions

Major: Astrophysics, Physics

Minor: Computer Science

EXPERIENCE

Machine Learning Intern, ASML (Veldhoven, Netherlands) October 2025 – June 2026

Working in the D&E department on machine learning solutions for diagnosing degradation in NXE lithography machines (various contaminations in the optical column).

- Researched hybrid models integrating physical simulations with machine learning, focusing on data-driven corrections to the physical models.
- Implemented Bayesian inference algorithms to infer contamination profiles in the machines based on the hybrid models. Methods used: Variational Inference, Normalizing flows, Variational Autoencoders.
- Developed high-performance implementations of inference algorithms, optimized for both edge devices (ordinary laptops) and large-scale HPC clusters.
- Resulting method is considered for the production use with estimated cost-savings on order of $> 20\text{m } \text{€}$ per year.

Quantitative Developer Intern, UBS AG

July 2025 – September 2025

Worked on a high-performance, multi-code C++ engine used to price loans within the Lombard Lending unit.

- Performed code benchmarking and bottleneck analysis to identify performance issues.
- Investigated parallel efficiency and load balancing using Intel TBB.
- Applied memory and computational optimizations, resulting in a codebase that was $3\times$ faster and $2\times$ more memory-efficient.

Student research assistant, Nicolaus Copernicus Astronomical Center April 2023 - July 2025

Working under supervision of Krzysztof Nalewajko (NCAC). Project in cooperation with Bart Ripperda (CITA, Toronto), Alexander Philippov (University of Maryland/Stanford University) and Agnieszka Janiuk (CFT, PAS).

- Performing basic research in the field of the high energy astrophysics (plasma physics, physics of accretion).
- Analyzing data from various HPC simulations starting from small scale local simulations up to global simulations.
- Scientific code development, most of my work was centered around HPC visualization software that was used to analyze multi TB snapshots from the simulations.
- Performed work was presented in two second-author publications.

Student research assistant at Astronomical Observatory July 2022 - July 2025
Student position in grant 2021/41/B/ST9/00252, working under the supervision of Przemysław Mróz.

- Performing basic research in the field of the observational astrophysics. Results include two first-author and one contributing-author publications.
- Scientific code development, main activities include writing custom Bayesian inference software based on MCMC methods, analyzing various observational data from various scientific surveys/telescopes including OGLE, Gaia, HST and more.
- Scientific topics: Microlensing, binary stars.

PROJECTS

Neural Posterior Estimation for Spectral Energy Distribution fitting Github

Researching applications of Neural Posterior Estimation in Astronomy. Spectral Energy Distribution fitting is powerful yet computationally expensive technique widely used in Astronomy. I demonstrated that with the help of normalizing flows one can speed up the inference of parameters by several orders of magnitudes. To do so, I designed my own complex preprocessing model that was paired with Masked Autoregressive Flow. Model was then fine-tuned so it can operate on real-life astronomical data. Model was accepted as poster at the ICML 2025 co-located workshop: Machine Learning for Astrophysics.

Computational Fluid Dynamics on GPU-s

Github

Astrophysical research heavily relies on the simulations. In the project, a Riemann solver was implemented to solve the equations of relativistic fluids. It is written in CUDA-enabled Julia code parallelized with MPI to utilize massive GPU clusters. It relies on manually written computational kernels that are optimized for A100 NVIDIA GPU-s (compute - memory bandwidth - communication). Project is being developed with the help of EuroHPC development access computing grant EHPC-DEV-2025D02-085 on the Leonardo BOOSTER, where I am a Co-PI.

TALKS & POSTERS

- **M. Kapusta:** "IrisML: Neural Posterior Estimation for the Spectral Energy Distribution fitting" ICML 2025: Machine learning for Astrophysics workshop.

PUBLICATIONS

- K. Nalewajko, **M. Kapusta**, B. Ripperda, A. Phillipov. "Distortion of a relativistic jet echoing a magnetic flux eruption" Submitted in Astrophysical Journal
- **M. Kapusta**, P. Mróz, et al. "HST pre-imaging of a free-floating planet candidate microlensing event" Accepted in Astronomy & Astrophysics [[arXiv:2507.01109](#)]
- K. Nalewajko, **M. Kapusta**, A. Janiuk "Chaotic Magnetic Disconnections Trigger Flux Eruptions in Accretion Flows Channeled onto Magnetically Saturated Kerr Black Holes" Accepted in Astronomy & Astrophysics [[arXiv:2410.08280](#)]
- P. Mróz, A. Udalski, M. Szymański, **M. Kapusta**, et al. "Microlensing Optical Depth and Event Rate toward the Large Magellanic Cloud Based on 20 yr of OGLE Observations" Accepted in ApJS [[arXiv:2403.02398](#)]
- **M. Kapusta**, P. Mróz. "The search for Dormant Black Holes in the OGLE data" Accepted in Acta Astronomica [[arXiv:2401.11293](#)]

AWARDS & SCHOLARSHIPS

- Gold medal at 2021 University Physics Competition (as part of the team representing Faculty of Physics)
- Silver medal at 1st Global e-Competition on Astronomy and Astrophysics (in place of 14th International Olympiad on Astronomy and Astrophysics), 2020

- Bronze medal at 13th International Olympiad on Astronomy and Astrophysics, 2019 Hungary
- Winner of 62th and 63th Polish Astronomy Olympiad
- Finalist of 67th and 69th (11th place) Polish Physics Olympiad
- Finalist of 70th Polish Mathematical Olympiad
- Minister of Education's scholarship in the year 2018/2019, 2019/2020
- Rector scholarship in the academic year 2020/2021, 2021/2022, 2022/2023, 2023/2024

OUTREACH

- Judge at International Math Competition Naboj (March, 2023-2025)
- Judge at 16th International Olympiad on Astronomy and Astrophysics (August, 2023)

COMPUTER SKILLS

Languages & Software:

- Computer languages - Python, C/C++, Julia, R, Rust, Fortran
- Deep learning knowledge using PyTorch/JAX.
- Parallel programming using MPI/OMP.
- GPU programming skills using CUDA (also paired with MPI).

Language

- English - C1 (103 TOEFL)
- Polish - native speaker
- German - A2/B1